

Building the GitHub Classroom Template Repo — exact steps

SYSC term project (JPacman)

JPacman (SERG-Delft/jpacman) is a **self-contained Gradle project** — no companion libraries, and static analysis (Checkstyle/PMD/SpotBugs) plus JaCoCo coverage are built in. So the template is just *JPacman + your classroom files*.

0. Prerequisites

JDK (match the JPacman Gradle wrapper — 11 is safe), Git, a GitHub account. Check: `java -version`, `git --version`.

1. Build JPacman once locally

```
git clone https://github.com/SERG-Delft/jpacman
cd jpacman
chmod +x ./gradlew
./gradlew build           # compiles, runs tests, JaCoCo report, Checkstyle/PMD/SpotBugs
./gradlew run             # optional: play a round
cd ..
```

The JaCoCo CSV lands at `build/reports/jacoco/test/jacocoTestReport.csv`.

2. Create the GitHub organization

GitHub → + (top right) → **New organization** → Free plan → name it (e.g. SYSC-4101).

3. Build the template repository

```
cp -r jpacman sysc-termproject-starter
cp -r /path/to/TermProject/ClassroomStarter/. sysc-termproject-starter/
cd sysc-termproject-starter
./gradlew build           # must succeed and produce the JaCoCo report
git add -A && git commit -m "SYSC term-project starter (JPacman + autograding)"
```

Keep JPacman's LICENSE (CC-BY-SA / MIT as applicable) and add an attribution line to README.md.

Push to the org (create an **empty** repo in the org first — New repository, no README):

```
git remote add origin https://github.com/<org>/sysc-termproject-starter.git
git branch -M main && git push -u origin main
```

4. Mark it a template + set the coverage scope

- Repo -> Settings -> check **Template repository**.

- **Settings** -> **Secrets and variables** -> **Actions** -> **Variables** -> **New repository variable**:
`COVERAGE_SCOPE = board,level,npc.`
- (No SonarCloud secret needed — static analysis is built into the Gradle build.)

5. Create the Classroom + roster

- <https://classroom.github.com> -> **New classroom** -> select your org -> name it.
- **Students** -> **Update roster** -> upload a CSV of student IDs.

6. Create the group assignment

- **New assignment** -> **Group assignment**; title; **Team size 3**; **Private**.
- **Starter code** -> **a template repository** -> `sysc-termproject-starter` -> **Create**.

7. Get the invite link

Assignment page -> **Copy invite link** (<https://classroom.github.com/a/...>) -> post on Brightspace.

Why this is simpler than a Maven multi-repo game: JPacman has no external companion libraries, so `./gradlew build` works out of the box — nothing to pre-install. The autograding workflows use `./gradlew` and the coverage checker reads the Gradle JaCoCo report.